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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B. Tech (CBCS) Degree Examinations

December 2024

EC1001-1 – BASIC ELECTRONICS
(FOR AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from Unit – I & Unit – II each and **One full** question from Unit – III.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Zener diode is mainly used as
 - A) Amplifier
 - B) Comparator
 - C) Oscillator
 - D) Voltage regulator
2. The r.m.s value of load current in a full wave rectifier is
 - A) $0.707 I_m$
 - B) 0.5π
 - C) $0.5 I_m$
 - D) $0.3183 I_m$
3. BJT is made to work as an amplifier when it is operated in region
 - A) Cut off
 - B) Active
 - C) Saturation
 - D) Inverted mode
4. In CE configuration of BJT, output characteristics is plotted by taking
 - A) V_{CE} verses I_C for constant V_{CB}
 - B) V_{CE} verses I_C for constant I_E
 - C) V_{CE} verses I_C for constant I_B
 - D) V_{CE} verse I_E for constant I_C
5. What is the state of an ideal diode in the region of non-conduction?
 - A) Short circuit
 - B) Open circuit
 - C) Undefined
 - D) Unpredictable
6. The collector to base current gain β in terms of α is defined as
 - A) $\frac{\alpha}{1 + \alpha}$
 - B) $\frac{\beta}{1 + \alpha}$
 - C) $\frac{\alpha}{1 - \alpha}$
 - D) $\frac{\beta}{1 + \beta}$
7. A germanium diode is connected in series to a dc supply of 4volts and resistance of 2 k Ω , the current in the circuit is
 - A) 1.45 mA
 - B) 1.85 mA
 - C) 0.185 mA
 - D) 18.5 mA
8. A bipolar junction transistor is a
 - A) Voltage controlled device
 - B) Resistance controlled device
 - C) Current controlled device
 - D) Junction controlled device
9. An integrator circuit using an Op Amp hasin its feedback path
 - A) Resistor
 - B) Capacitor
 - C) Inductor
 - D) Diode
10. In Colpitts' oscillator, the components used in the feedback network are
 - A) Two Inductor and one Capacitors
 - B) Two resistors and two capacitors
 - C) Two Capacitors and one Inductor
 - D) Two inductors and two capacitors
11. With a resistance value of $R=1k\Omega$ in a feedback network of RC oscillator, frequency of oscillations generated is 5 kHz. The value of the capacitor C is
 - A) $0.129 \mu F$
 - B) $0.0219 \mu F$
 - C) $129 \mu F$
 - D) $0.0129 \mu F$

12. The outputs of two comparators are connected to a in the internal circuitry of IC 555 timer
 A) J-K Flip Flop B) S-R Flip Flop
 C) D-type Flip Flop D) T-Type Flip Flop
13. In an amplifier, positive feedback leads to
 A) Amplification B) Breakdown
 C) Avalanche-breakdown D) Oscillations
14. Which is not the ideal characteristic of an op-amp?
 A) Output impedance is zero B) Input Resistance is zero
 C) Bandwidth is infinity D) Open loop voltage gain is infinity
15. Which of the following is not the advantages of negative feedback amplifier?
 A) Higher input impedance B) Reduction in noise
 C) Unstable gain D) Lower output impedance
16. In an RC phase shift oscillator, the resistances in the feedback network are 4.7 kΩ and capacitor values are C = 0.47 μF. The frequency of oscillations is
 A) 294.13 Hz B) 29.413 KHz
 C) 2941.3 Hz D) 29.413 Hz
17. The initial mobile communication systems for public safety in United States used the following modulation technique,
 A) Ampere modulation B) Frequency modulation
 C) Amplitude modulation D) Phase modulation
18. The mechanism of using the same frequency band within a geographical area in a Cellular or mobile communication system is referred to as,
 A) Efficiency B) Frequency reuse
 C) Reliability D) Bandwidth coordination
19. On which component modulation is performed?
 A) Receiver B) Sensor
 C) Transmitter D) Actuator
20. The audio frequency range is
 A) 20 kHz to 200 KHz B) 20 Hz to 20 KHz
 C) 2 Hz to 20 MHz D) 30 MHz to 200 MHz

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With the help of neat diagrams explain the ideal and piecewise pn junction Silicon diode models.	5	L*2	1	1
	b) Show that the amount of ac present in the output of the dc voltage of a full wave rectifier is 48.2 %.	6	L3	1	1
	c) A bridge rectifier uses four diodes with rms voltage of 110v. The forward resistance of each diode is 25Ω, with load resistance of 500Ω. Find (i) maximum value of I_m current, (ii) DC value of current through load and (iii) DC value of voltage across the load resistance.	5	L3	1	1
2.	a) With a neat circuit diagram, explain how BJT can be made to operate as a switch.	5	L2	2	1
	b) Define with suitable equations, line regulation and load regulation in Zener voltage regulators.	6	L2	1	1
	c) Calculate α and β of a transistor, taking collector current as 1mA and base current as 25μA. Also determine the new base current to give collector current as 5mA.	5	L3	2	1

3. a) For the diode circuit shown in figure Q3(a), calculate the current I and the output voltage V_O .

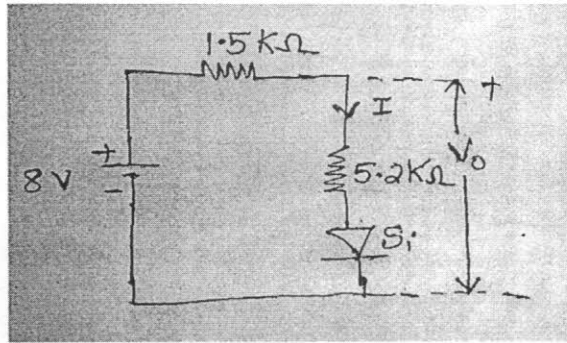


Fig. Q3(a)

- b) Draw and describe the significance of each component in the electronic circuit for a common emitter-resistor and capacitor coupled amplifier using NPN BJT transistor.
- c) Explain the working of a CMOS inverter with a neat circuit diagram.

5	L3	2	1
6	L2	2	1
5	L2	2	1

Unit – II

4. a) List and mention corresponding values of the ideal characteristics of an Op-Amp.
- b) Explain with neat circuit diagram and relevant waveforms, the operation of non-inverting comparator with positive reference voltage.
- c) Write the name of the Op-amp circuit shown in figure Q4(c). If the gain of the Op-amp is 60, find the value of the resistor R_f .

5	L2	3	1
6	L2	3	1

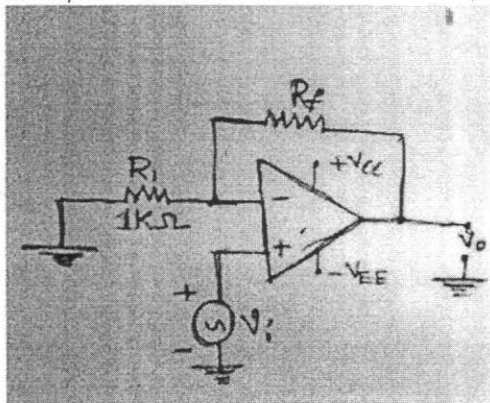


Fig. 4Q(c)

5. a) In a non-inverting amplifier using Op-amp $R_1=2K\Omega$, $R_f=200K\Omega$. Supply voltages are $\pm 15V$. Calculate V_O if the applied input voltage is $1.5V_{PP}$. Draw the input and output waveforms.
- b) With a neat circuit diagram, explain the operation of a Colpitts' oscillator. Write the equation of frequency of oscillation.
- c) List out the advantages of negative feedback.
6. a) For an IC timer based astable multivibrator Duty cycle is 75%, frequency is 1KHz, $R_2=3.6K\Omega$ and $C=0.1\mu F$. Calculate T_{ON} and R_1 .
- b) Discuss the concept of voltage series negative feedback with a neat block diagram.
- c) State and explain Barkhausen's criterion for generating sustained oscillations.

5	L3	3	1
5	L2	3	1
6	L2	4	1
5	L3	4	1
5	L3	3	1
6	L2	4	1
5	L2	4	1

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Write any five comparisons between microprocessor and microcontroller. | 5 | L2 | 5 | 1 |
| b) | With neat block diagram discuss the basic elements of communication systems. | 6 | L2 | 5 | 1 |
| c) | Draw the diagram of a cellular system and show the frequency reuse mechanism in it. | 5 | L2 | 5 | 1 |
| 8. a) | Describe the types of communication channels used for the transmission of signals. | 5 | L2 | 5 | 1 |
| b) | What is sensor and actuator? Explain how the LED and opto-coupler are used as an I/O subsystem in embedded system. | 6 | L2 | 5 | 1 |
| c) | With a neat block diagram, explain the elements of an embedded systems. | 5 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
